



Understanding the experience of neurodivergent workers in image and text data annotation

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ABSTRACT

With the rise of large-scale data-driven innovation in AI, data annotation tasks found in digital work environments present an employment opportunity for neurodivergent individuals. Though work in data annotation can potentially ease the high unemployment rate of neurodivergent individuals, limited research focuses on the experience of neurodivergent workers in data annotation micro-tasks. To aid in understanding the experience of neurodivergent crowd workers, we conducted a user study with ten neurodivergent workers between the ages of 18–30. Participants completed three types of micro-tasks in a custom web-based data annotation platform. With the data collected from the platform, we examined individual responses within data annotations, work completion, the time to complete work, and calculated a potential “effective hourly wage,” for each participant based on their responses. Through a survey and semi-structured interview following each task, we learned about the experience of all participants regarding each of the data annotation tasks. Results of the study show: 1) our participants provide diverse annotations that are valuable for employers in digital data annotation work environments; 2) when calculating the “effective hourly wage” of all participants per task, some of our participants would earn less than minimum hourly wage on tasks; and 3) participant perceptions of the tasks matched their responses in the tasks presented.

1. Introduction

Neurodivergent individuals, such as those diagnosed with autism spectrum disorder (ASD) and attention deficit and hyperactivity disorder (ADHD), face numerous challenges associated with participation in the workforce. Data from the National Longitudinal Transition Study-2 (NLTS-2), the largest comprehensive post-graduation survey of youth with disabilities in the U.S., showed that only 37% of young adults with ASD within the study were employed up to eight years after graduating from high school and that 34.9% reported never holding a job since high school (Shattuck et al., 2012; Wagner et al., 2006). It is important to highlight that this study did not include any breakdown by ASD severity and any comorbidities, so it is likely that for some segments of the ASD population the statistics are even more discouraging. A systematic review of research on the transition of youth diagnosed with ADHD (Gordon & Fabiano, 2019), also found that teenagers and young adults with ADHD had lower occupational attainment and demonstrated job

instability. These difficulties for neurodivergent individuals to obtain and maintain employment may be due to challenges in obtaining transportation, challenges with communication, relating to others' experience, and maintaining flexibility and focus with rapid changes in tasks presented in the workplace (Schall & McDonough, 2010; Schall et al., 2020; Woods et al., 2002; Yaneva et al., 2015; Tomczak & Kulikowski, 2023).

To address these challenges, crowd work provides an interesting opportunity for neurodivergent workers. Crowd work often involves data annotation in micro-tasks, such as those found within Amazon's Mechanical Turk (AMT) within a remote and digital work environment, where the “worker” can work from the comfort of their own home and choose tasks to complete for payment. These digital work environments may provide a path to employment and highlight the strengths of neurodivergent workers (Hara & Bigham, 2017, October; Zyskowski et al., 2015). With an increased emphasis on building fair AI models (Gao et al., 2022; Park et al., 2021), particularly in reducing the bias that

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currently exists for neurodivergent individuals (Guo et al., 2020; You et al., 2022), it is also important to obtain datasets annotated by neurodivergent workers to aid in building fair, inclusive AI systems. To study this, however, we must first learn from the experience of neurodivergent workers on data annotation tasks within these digital work environments.

Because little academic research addresses neurodiverse workers (LeFevre-Levy et al., 2023), an inductive approach may be an appropriate beginning. The objective of our exploratory study was to gain insights into the feasibility of data annotation work for neurodivergent workers. We studied the feasibility along two dimensions—the task performance and perspectives of participants. To study these dimensions, we recruited ten self-identified neurodivergent participants who were 18–30 years of age. We built a custom web platform that presented 3 types of text and image annotation tasks with 20 questions for each task. These were tasks that were common representations of objective and subjective tasks typically found in platforms such as AMT. Upon completion of each task, we conducted semi-structured interviews and a survey to gain insight into the experience of neurodivergent workers regarding the tasks presented in the platform. In addition to the interview, we also encouraged our users to “talk aloud” during the completion of the tasks to share their experience interacting with the interface and completing tasks.

During our study, we faced challenges in gathering information from our participants despite our efforts to present multimodal questions during the interviews. We found that neurodivergent participants did not prefer to “talk aloud” while working on the tasks. From this experience, we suggest that researchers be mindful of different communication preferences, such as vocal or written communication, when interviewing neurodivergent participants. Based on the observations of individual task completion and the experience of our participants, we provide suggestions for potential employers of neurodivergent workers regarding the design of tasks in digital work environments.

2. Related work

There is a small but growing body of research studying how neurodivergent workers experience crowd work (Hara & Bigham, 2017, October; Hara et al., 2019; Uzor et al., 2021; Zyskowski et al., 2015). Crowd work, specifically in work involving data annotation, often involves significant attention to detail and detail processing. Some neurodivergent workers may have strengths in memory ability, information processing, and may have a strong interest in digital technology (Murray, 2018). Tomczak and Kulikowski (2023) discuss these strengths, citing research from authors that specifically evaluated perspectives of those with ASD. They propose that compared to neurotypical people, if work is paired with these strengths, it could reduce burnout (Tomczak & Kulikowski, 2023). Additionally, in an interview with crowd workers and job coaches with disabilities, which included a subgroup that identified as neurodivergent, Zyskowski et al. (2015) found that both job coaches and workers regard crowd work as a viable source of employment and that neurodivergent workers currently participate as remote crowd workers.

Zyskowski (2015) also highlighted the concerns of workers regarding pay in crowd work settings. The workers reported that their main concerns with crowd work were user design, pay, and performance ratings, and, to a lesser extent, the lack of socialization opportunities (Zyskowski et al., 2015). It was estimated that most crowd workers make less than minimum wage (Hara et al., 2018, 2019; Hornuf & Vrankar, 2022), and neurodivergent workers are often underemployed, which leads to fewer chances for financial independence (Khalifa et al., 2020). Additional studies analyzed payment issues within the micro-task economy, noticing that crowd workers often spend a significant amount of time working on tasks without pay – searching for tasks, working on rejected tasks, or starting work without completing tasks (Uzor et al., 2021). Uzor et al. (2021) completed a survey of 600 participants identified as having

various disabilities to gain insight into the accessibility challenges of crowd work platforms such as Amazon Mechanical Turk. They observed that 43% of the participants struggled with the time required to complete micro-tasks, there was insufficient time to complete the tasks to their satisfaction, and some participants reported higher levels of anxiety and depression due to these time limits.

The challenges identified in these previous studies highlight the pressing need for changes within digital work platforms for data annotation. However, when developing these digital work environments to be favorable for neurodivergent workers in employment, we must also understand the experience of neurodivergent workers—their responses and perspectives regarding data annotation tasks (Praslova et al., 2023). Given the potential of these digital work environments as an avenue to reduce unemployment in the neurodivergent population, our study aims to provide insights into the opportunities and obstacles in reaching that goal, such as identifying task challenges and suggestions for digital work features.

3. Methods

For the study, we recruited neurodivergent participants, and built a custom web platform that presented 3 types of text and image annotation tasks that were common representations of objective and subjective tasks typically found in platforms such as AMT. Upon the completion of each task, we conducted a semi-structured interview and survey to gain insight into the experience of our participants regarding the tasks presented within the platform. This section outlines the methodology of our exploratory study.

3.1. Participants

We recruited 10 self-identified neurodivergent participants between the ages of 18–30, through social media postings and emails to local agencies supporting neurodivergent individuals in employment (Table 2). Following recruitment, we asked participants to take part in an informed consent meeting to explain the study and answer any questions they had about their participation. All screened participants consented to participate. An Institutional Review Board approved all procedures.

3.2. Custom platform and tasks

We chose to build a custom platform to tailor data collection and perform our study remotely through Zoom using features such as the share screen and the participant control screen function in the Zoom platform. This was for two reasons. First, we were able to log a wider range of worker behavior than is possible using AMT, such as recording response latency, and participant duration per question. Second, using our platform, we could interact with the workers remotely through Zoom and could record the work sessions for analysis, which is not enabled by AMT. AMT restrictions prevented performing such a user study on the AMT platform using Zoom. Remote interaction with Zoom was a necessity, because due to COVID-19 restrictions, we were not able to have in person contact with participants. We created the tasks to closely resemble common tasks at AMT and followed previous research suggestions in design regarding the task design (Pavlov, 2014; Sarhan, 2019) and task instructions (Manam and Quinn, 2018). Within each task, we also implemented a “Skip” feature that allowed users to skip questions and explain their reasons for skipping a question. The platform measured time spent on a task and logged cursor movements, keyboard strokes, response latency, and mouse button clicks. We used a Latin square design (Gao, 2005) for the presentation of tasks and randomly ordered our questions within each task, to reduce the likelihood of ordering effects on participant responses (Cai et al., 2016). Access to the platform will be available upon request.

We studied three different tasks: receipt transcription, picture

analysis, and movie reviews. These tasks were representative of some image and text annotation tasks found in AMT, with 20 questions per task, randomly selected from each of the datasets per task in our custom platform. We also designed our tasks to be similar to those used in previous research (Hara & Bigham, 2017, October). The summary of the tasks is shown in Table 1.

3.3. Data collection

Following the consent meeting, participants met with the researchers for separate “work sessions” via Zoom. We gave participants access to the web platform through a computer dedicated to the study in the research lab and Zoom’s remote function. With participant consent, all sessions were recorded and transcribed by Zoom. Some participants required more time to complete tasks and were asked to return for a second and even third “work sessions” to complete the tasks. Most sessions lasted around 60 min. Following each task during each “work session,” participants were asked if they needed time for a break.

Each task started with two practice questions. We used the first question to model for the participant how to use the web platform to respond to questions, as well as how to use the skip and submit buttons featured on each question page. The second question allowed the participant to practice. We provided help only if the participant had issues or questions about interacting with the platform. As this was an exploratory study, to study the participants’ behavior and experiences, we did not provide direct feedback on their responses in the platform while answering questions. We also modeled for participants how to “talk-aloud” (Laffal et al., 1985) while working on the tasks. This was in the form of narrating the selection of responses and why they were selecting the response.

Following the two practice questions, participants worked independently on the remaining 18 questions. During this time, we muted the microphone and turned the camera off. We designed this protocol to measure how participants interacted with the tasks. If participants asked questions during the tasks, we unmuted the microphone and turned on the video camera to respond. If the participant asked for feedback about their responses, such as if they were correct or incorrect, we responded with “Do your best,” and did not provide feedback on the response. If the participant asked a technical question (e.g., mouse battery needed changing), or for a break, we responded to their question.

Following each task, we asked participants questions from NASA-TLX, a reliable and valid tool that measures subjective workload assessment in tasks (Hart, 2006; Xiao et al., 2005). Participants were asked to rate on a scale of one through five (one being strongly disagree

and five being strongly agree) the following questions: “Was the task complex?” (Mental work), “Was the physical activity strenuous?” (Physical work), “Did you work hard to accomplish your level of responding” (Effort), “Did you feel your pace completing the task was fast?” (Pace), “Did you feel successful?” (Success), “Did you feel frustrated?” (Frustration). We asked each participant why they selected their response to the survey questions, and we also asked additional open-ended interview questions to gather more information regarding the participant’s experience, such as “Did you experience any challenges, and if yes, what were they?”, “Is there anything you would want to change about the tasks?”, and “Is there anything in addition that you would like to share about your experience while completing the tasks?”

3.4. Data analysis

To answer research questions about task completion and the experience of the participants, we used a different data analysis approach for each task, as each task presented varied data analysis challenges. For tasks where we had a ground truth annotation (receipt transcription and movie review), we calculated accuracy and mean absolute error. For subjective annotation tasks (picture description), we classified responses into common categories, following suggestions for subjective data annotation analysis from previous research (Alm, 2011; Hube et al., 2019). To understand the experience of our participants, we collected data from the survey and open-ended interview questions from the NASA-TLX (Hart, 2006; Hong et al., 2018; Yan et al., 2022).

4. Analysis and results

4.1. Analysis of responses

Receipt transcription. For the receipt transcription task, we determined the ground truth answers by examining each receipt. We then manually compared the answers provided by each participant with the ground truth. We ignored questions that participants skipped when using the skip button. For each participant, we calculated the percentage of correct entries in text boxes. We deemed the answer incorrect if the answer was different from the ground truth, ignoring minor spelling mistakes and capitalization.

The Receipt column in Table 2 reports the accuracy for each participant. One participant reached an accuracy of 96%. Six participants had an accuracy over 90%. We noticed that some participants made consistent errors, such as entering incomplete street names (P02).

Movie review. The dataset contained ground truth labels (on a scale of 1–10; with 1 being bad and 10 being excellent) about the movie review. We calculated the mean absolute error (MAE) as the average absolute difference between the response and ground truth. Bias scores were calculated as the average difference between the responses and ground truth.

We observe from the MAE column in Table 2 seven of the participants (all self-identified as having ASD) had scores greater than a MAE of 2.0. When examining the properties for this task, we find that this task required attention, focus, and emotion recognition, which may be challenging for individuals with ASD (Baron-Cohen, 2000; Brewer et al., 2016; Schall & McDonough, 2010). All three participants with MAE less than 2.0 self-identified with ADHD. Though they have challenges in attention and focus, individuals with ADHD do not necessarily experience challenges with emotion recognition (Woods et al., 2002).

We also asked participants to explain why they selected each sentiment to understand their reasoning behind the answers. As shown in Table 2, we coded the explanations into three categories: 1) explaining the tweet (Desc), 2) out of context (OOC; off topic or did not match the context of the tweet), and 3) personal association (Assoc). An example of a personal association is the comment “I love the Guardians of the Galaxy movie,” when justifying their rating selection that mentioned the movie “Guardians of the Galaxy.” Six out of seven participants with ASD

Table 1
Summary of tasks.

Task	Dataset	Description
Receipt Transcription	ExpressExpense Sample Receipt Database (Janfk, n. d).	Participants were asked to fill out the information for the shown receipt from an image of a restaurant receipt
Movie Reviews	IMDB movie review dataset (Maas et al., 2011),	Participants were asked to: “guess the rating that the reviewer entered (1 = Bad, 10 = Great)” by selecting the number of stars. In addition, participants were asked “why did you choose this rating” to provide insight into their answers.
Picture Description	Visual Genome (Krishna et al., 2017).	Participants were asked to “write five descriptive words for the image” and “write a sentence to describe the image using at least 10 words.” Five separate text boxes were provided for the keywords and one text box for the sentence.

Table 2

a) Participant id, b) participant self-identified diagnosis, c) total skips, d) average duration in seconds per question per task, e) mean accuracy for receipt transcription, f) mean absolute error movie review tasks, g) mean bias per participant in movie reviews, and h-j) and justification categories for movie review tasks (Description, Out of Context, and Association).

Id	Dx	Skips	Duration	Receipt	Movie				
				Accuracy	MAE	Bias	Desc	OOC	Assoc
P01	ASD	9	94.2	92%	3.5	-1.28	0	3	15
P02	ASD	1	78.6	78%	2.72	1.28	6	8	4
P03	ASD	2	62.2	94%	2.06	0.28	17	0	0
P04	ADHD	4	68.2	96%	1.17	0.61	18	0	0
P05	ASD, ADHD	0	130.4	93%	1.39	-0.06	18	0	0
P06	ASD	7	204.0	89%	3.89	0.11	3	9	2
P07	ASD	9	105.6	74%	2.94	-2.39	9	0	4
P08	ADHD	1	68.5	91%	1.39	-1.06	15	0	3
P09	ASD	22	61.7	61%	3.5	0.94	1	15	2
P10	ASD	0	79.7	93%	2.06	0.06	14	0	4
Average		5.5	95.3	86%	2.46	-0.15	10.1	3.5	3.4

(P01, P02, P06, and P09) often used out of context or personal associations for their justifications. It is important to note that this was a subjective task, and these responses are not necessarily “incorrect” but were different and not directly related to the movie review.

Picture Description. In this task, despite the Visual Genome dataset containing a specific set of attributes and object labels per picture, questions to the participants were open ended. To analyze the five keywords entered to describe each picture, we manually categorized the entered keywords into six coding classes: 1) objects (e.g. labeling actual items in the picture), 2) setting (e.g. labeling where the picture appeared to be located, such as a “city” or “forest”), 3) actions (e.g. labeling what someone was doing in the picture), 4) emotions (e.g. labeling how someone may be feeling such as “happy,” or “sad”), 5) attributes (e.g. labeling colors, shapes, or textures), 6) associations (e.g. labeling a personal connection to the picture, such as the name of something they were reminded of in the image, but not present in the picture).

We counted the distribution of keyword categories for participants as shown in Fig. 1. Overall, the most common keywords were in the object category, followed by setting, attributes, and actions. Emotion and associations were used very rarely. Some of the associative annotations entered by participants related to their own personal experience rather than a description of the picture. For example, one participant entered the name of the resort they visited when shown a beach picture.

4.2. Analysis of work completion

Work completion is important in data annotation, as skipped responses lead to missing data, which is undesirable for employers in data

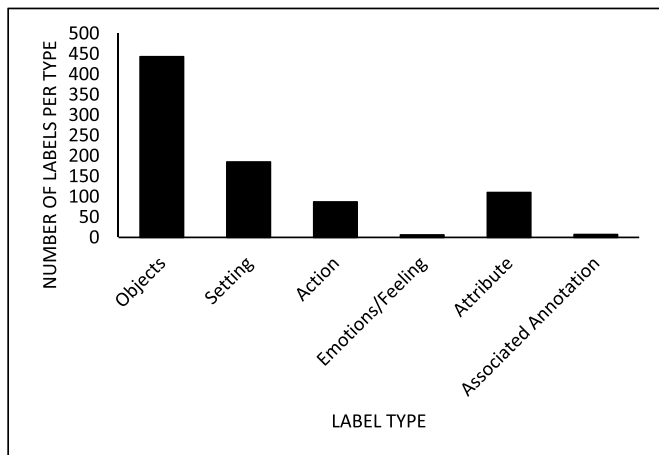


Fig. 1. Distribution of number of label types for the picture description task.

annotation work. We measured work completion by counting how often participants did not answer parts of questions or when they skipped whole questions. As shown in the Skip column in Table 2, four participants had 5 or more skipped questions, and these participants self-identified as having ASD. We observed that the picture description and movie review tasks caused the largest number of incomplete or skipped questions. These tasks required the participant to read long passages, and type novel responses based on what was provided, as compared to the receipt transcription task which involved transcribing what was in the image.

4.3. Analysis of time per question

Speed is an important component of work. Crowd workers are typically paid a flat fee to complete a task, and the time to complete a task directly influences the hourly rate for crowd work. The time column in Table 3 shows the mean time in seconds per question across all tasks. P05 and P06 took the longest per question (Table 2), and we noticed that they spent more time with movie reviews which involved reading long paragraphs of text. Upon inspection of Zoom sessions, we observed that P05 was working on a second screen in parallel with our tasks (presumably on other activities) which may have affected their time per question.

4.4. Analysis of effective hourly pay

Worker payment for data annotation tasks, as previous research notes, is a major issue in crowdwork (Hara et al., 2018; Uzor et al., 2021). A study by Hara et al., in 2018 estimated that the hourly wage for crowd workers on AMT ranged from \$4.80 to \$6.19 per hour (Hara et al., 2018). In this section we wanted to compare the pay potential for

Table 3

Effective hourly wage per task.

Id	Movies		Receipts	Average per participant
	Picture Analysis			
P01	\$5.52	\$7.83	\$18.38	\$10.58
P02	\$12.50	\$8.26	\$9.60	\$10.12
P03	\$12.75	\$12.61	\$25.64	\$17.00
P04	\$10.04	\$14.02	\$22.34	\$15.47
P05	\$5.33	\$7.99	\$8.08	\$7.13
P06	\$2.05	\$9.24	\$4.86	\$5.38
P07	\$14.39	\$11.43	\$3.69	\$9.83
P08	\$12.77	\$9.86	\$18.93	\$13.86
P09	\$15.22	\$1.95	\$11.37	\$9.51
P10	\$11.26	\$12.66	\$7.82	\$10.58
Average per task	\$10.18	\$9.59	\$13.07	

participants in our study. To make it easier to follow the results, we assumed that the hourly pay (HP) of an average successful worker in our participant group is \$10 per hour ($HP = \10).¹ Let us denote as t the time a particular worker needs to complete a task. We define a successful worker as a worker who answered all questions in a particular task, regardless of the time needed for them to answer the questions and regardless of the accuracy of the answers. Let us denote as $\text{avg}(t_{EX})$ as the average time successful workers needed to complete a task. We calculate the effective hourly pay (EHP) for a particular worker on a task as $EHP = HP \cdot \text{avg}(t_{EX})/t$. For example, if the average time on a task of successful workers was 10 min, and a particular successful worker needed 20 min to complete it, that worker's EHP would be \$5 per hour.

However, because our platform allows workers to skip questions (and this is a common feature on most crowdsourcing platforms), we assumed that such workers should not be paid for skipped questions. Thus, we calculated the fraction of answered questions on a task f for each worker ($f = 1$ if the worker answered all the questions). Then, the EHP is calculated as $EHP = HP \cdot f \cdot \text{avg}(t_{EX})/t$. For example, if the worker from the previous example who spent 20 min on the task answered only 50% of the questions, that worker's EHP would be only \$2.50 per hour.

From this, we calculated the EHP for all participants in our study on each of the three tasks and show the results in Table 3. We observe that the average EHP for our participant group was \$10.18 for the movie task, \$9.59 for the picture task (less than our set hourly pay); and \$13.07 for the receipt task. The lowest paid worker (P09) on the picture analysis task, would earn only \$1.95.

It is also worth noting that task requesters for crowd work are likely to refuse payment if the answer quality is not satisfactory. In addition, they might even refuse to pay if there are many skipped questions or if the work is not finished fast enough. Given this, we observe that the participants that skipped many questions (i.e. P06 and P09) would risk not getting paid at all for their work.

4.5. Analysis of participant experience

To gain insight on participant experience regarding the tasks, we asked participants to rate six NASA-TLX questions (Mental work, Physical work, Effort, Pace, Success, Frustration) (Hart, 2006) on a 5-point Likert scale. When comparing individual responses and their experiences, we observed some interesting insights. P02, P06, and P10 rated effort as being very high (giving it 5 ratings). It is noteworthy that P02 and P10 never skipped more than one question, while P06 spent the longest time on the tasks. P09 reported the highest level of frustration across tasks (average of 3, compared to the other participants' average of 1) and this participant had the most skipped questions.

We also asked participants open-ended interview questions regarding their experience as they were going through the tasks, and encouraged them to "talk aloud," while going through the tasks. Despite having a model of "talk aloud" at the beginning of each task, participants chose not to talk aloud during their work on tasks. This is consistent with the previous literature stating neurodivergent participants, particularly those with ASD, may have difficulty in with vocal communication (Schall & McDonough, 2010). Thus, gathering qualitative information using this method presented a challenge.

When asking our open-ended interview questions of participants, we provided the option for participants to type in Zoom's chat feature or vocalize their responses to these questions, and we found that two participants, P02 and P09, preferred to type some of their responses. In reviewing the transcripts from interviews, we can group the responses from participants into three themes regarding the tasks: task preferences, challenges with tasks, and suggested features for digital work environments and tasks.

¹ In Pennsylvania, where the study was conducted, minimum wage per hour is set at \$7.25 per hour.

4.5.1. Task preferences

Overall, most participants found the tasks to be enjoyable. Regarding the receipt task, participants found this task to be easy because: "It was very straightforward (P04)," and "I was really good at this because the information for the addresses and phone numbers were right there (P10)." For the picture analysis task, P07 noted, "It was easy because I could easily find things in the picture. It's a fun assignment to get paid for," and P05 stated, "There's really no wrong answers unless you're just describing something that's not there." Several participants enjoyed the movie review task because, "movies make me happy (P06)," and "that's my thing. I love movies! It was fun! (P08)." It was interesting to also observe that P03 and P04 both chose to write sentences before providing the descriptive words for the picture analysis task, working directionally from the bottom to the top of the page. When asked about this preference, P04 stated: "Creating the sentence is easy to do first because you are making an outline for yourself to come up with the five words."

4.5.2. Challenges with tasks

All participants reported challenges with deciding annotations for the subjective tasks. For the picture analysis task, P01 stated: "It's kind of hard to decide on a descriptive word. Some of the images do not have many details," and P05 stated: "Some pictures I felt like I had a lot to work with in terms of describing it, and then there were some where it felt like I couldn't really say a lot without repeating myself." For the movie review task, P02 stated that "I worked slowly because I was reading the sentences. It took a while. These questions were hard," while P03 reported that: "It was hard to determine the rating based on what they said." P07 also reported that working alone on the tasks without the researcher present caused some anxiety. They said: "My anxiety takes over when I have to work by myself. What if I accidentally type in the wrong information?" For receipt transcription, participants reported challenges with handling missing information: "Typing and finding the state on the receipt was hard. It was also hard when the phone number wasn't there." We did provide participants with written instructions to skip missing information; however, participants still reported this as a challenge. In addition to handling missing information on the receipt transcription task, participants reported trouble reading the receipts. P05 reported: "Some of them were small and hard to read from the screen."

4.5.3. Suggested features for digital work environments

Participants also provided suggestions for setting up inclusive and accessible digital work environments in data annotation for neurodivergent workers. Participants suggested that requesters provide clear instructions for tasks with visual examples, visual and auditory stimuli, accessible keyboard functions, word counters, and screen readers. For auditory and visual stimuli, participants requested auditory and visual stimuli to provide feedback and backgrounds on the screen to help them focus. P05 stated, "It would be nice to have something to help focus after each question, either a sound or an image." Specifically in the movie task where participants had to read long passages of text, P08 stated "The reading kind slowed me down; it would be good to have something to read the text for me."

5. Discussion and conclusion

Based on the responses and experience of participants in this study, our findings have implications for designing tasks in digital work environments that highlight the strengths of neurodivergent workers. We found that in our study, participants had a higher quality of responses on objective tasks such as receipt transcription. Thus, digital work environments containing more objective tasks may be preferable for some neurodivergent workers. It is important to note, however, that neurodivergent workers may not effectively identify preferable tasks unless they have access to the needed information to make these decisions. Previous related research developed a browser plug-in that aggregates data from AMT, enabling crowd workers to self-identify tasks that

yielded higher wages (Callison-Burch, 2014), thus reducing the penalty that “invisible labor” (Toxtli et al., 2021) may pose for neurodivergent workers. Our findings from the use of our custom platform, could inform the development of similar tools within real world data annotation platforms to enable neurodivergent workers to select tasks that best match their skill sets and are more likely to yield higher wages and increase job satisfaction.

Incorporating individual preferences and matching unique skill sets to the task can also help workers perform better and decrease pay discrepancies caused by skipped or incomplete responses (Uzor et al., 2021; Walsh et al., 2020; Zyskowski et al., 2015). This would also benefit requesters or employers of such tasks; in that they would get higher quality data annotations. Though we echo previous findings that worker-job matching while providing assistive, inclusive technology is important when recruiting neurodivergent participants for tasks in digital work environments (Hara & Bigham, 2017, October; Tavanapour & Bittner, 2019), further study is needed to evaluate employer perspectives regarding incorporating task preference, designing tasks and task instructions within digital work environments. In our study, we designed our tasks within our custom platform with instructions based on suggestions in design regarding the task design (Pavlov, 2014; Sarhan, 2019) and task instructions (Manam and Quinn, 2018). Task requesters on platforms such as AMT often post tasks with unclear or ambiguous directions (Manam and Quinn, 2018), which may frustrate or confuse both neurodivergent and non-neurodivergent workers. Developing assistive and inclusive tools to support in the learning of these tasks and reducing frustration may be beneficial for all data annotation workers.

Based on our observations on the individual responses of our participants within our platform, it is possible to envision building customized virtual coaching and support systems for some participants within current data annotation platforms such as AMT, perhaps through the use of generative AI language and computer vision models. For one participant (P09) we observed that they skipped many questions within the tasks. Mechanisms such as auditory or visual prompts built into the digital work platform may reduce skipped responses for this participant. While such assistive tools are difficult to add to AMT, they may be implemented through browser plug-ins. Companies that provide in-house labeling on custom platforms could also integrate these tools into their current systems. Participants P06 and P07 required more time to complete tasks. Matching these participants to tasks that do not require timed responses would prevent penalties (loss of wages) for the slower pace. Regarding working independently, in a digital work environment setting, P05 stated: *“I did find myself putting something on to listen to on my other monitor here. Because I don’t like it when there’s just silence, and I have no external stimuli when doing repetitive tasks like this.”* Providing a built-in system with background music or visual stimuli for this worker may be beneficial. On the other hand, related research found that too many stimuli may distract some workers in a digital work environment (Das et al., 2021). Thus, such environments need to be customized to individual worker preferences.

In addition to developing tools for support, *researchers should be aware of the preferred communication methods of neurodivergent study participants*. For example, we observed for two of our participants, typing into a text box was preferred in comparison to vocal communication. By incorporating the preferred communication method by neurodivergent participants, as researchers, we were able to gather more information on participant experience. This would, in turn, facilitate the inclusion of diverse viewpoints in design studies. We echo the statements of previous researchers supporting the inclusion of neurodivergent participants on research teams, so that study design is better informed by their experience (Motti, 2019; Sam & Hillman, 2022; Spiel et al., 2020).

Another critical challenge for future research is *to understand if crowd work can be an entry point for long-term positions*, such as work in data entry, medical billing, and office administration. While many tools have been developed for workers who are blind, or who have low vision (Kacorri et al., 2017; Medina et al., 2022), very few studies have focused

on developing tools for neurodivergent workers (Motti, 2019; Rapp et al., 2019). There has been some research on framing micro-tasks as internship opportunities for crowd workers (Suzuki et al., 2016). However, more research is needed in this space to study the link between micro-tasks in digital work environments and long-term employment outcomes. Future studies could investigate training neurodivergent workers using real world data annotation platforms such as AMT and evaluating if the skills in training generalize to other data entry settings in accounting and medical offices.

An obvious limitation was the small sample size of the study. With ten participants, it was hard to generalize our observation of responses to a larger neurodivergent population on data annotation tasks, especially when considering the diverse subgroups of our participants, where seven participants self-identified as having an ASD, and two self-identified as having ADHD. One participant identified as having both ASD and ADHD. For this reason, we did not have enough statistical power to make general conclusions about ASD and ADHD subgroups. Another limitation is that this paper targeted neurodivergent participants rather than neurodivergent and non-neurodivergent participants. The purpose of the study was to highlight the experiences of neurodivergent workers when completing data annotation tasks, rather than a comparison of their responses to a different group of participants. Regardless, we believe that these results provide helpful information that may inform follow-up studies, such as an experimental comparison of neurodivergent and non-neurodivergent worker responses on “real-world” tasks found in AMT or other platforms. Future studies could investigate the similarities and differences between subgroups of neurodivergent workers and responses in data annotation tasks.

It is important to note that another limitation is in the types of tasks selected for the study. For this study, we evaluated three different tasks, that were inspired from real world AMT tasks. These three tasks were chosen to provide a representation of various image and text annotation tasks, but there are many more types of tasks on platforms such as AMT, such as image annotation, object recognition, and audio transcription tasks. Future studies should evaluate the responses and experiences of neurodivergent workers across all of the different types of tasks found in platforms such as AMT, in order to gain more insight into task preferences and challenges.

In conclusion, the results from this exploratory study provide some insight regarding task completion, effective pay, and the experiences of neurodivergent workers when completing image and text-based data annotation tasks. These results suggest that some micro-tasks may be a good match for the unique skill sets of some neurodivergent workers. As a byproduct of our results, we also observed potential issues with wages for our participants in different micro-tasks. Based on this information, our team is working on future studies evaluating the task design and instruction preferences of neurodivergent workers, as well as studies to develop and evaluate generative AI prototypes that provide individualized coaching and support in data annotation tasks within real world digital work environments. These future studies may reduce wage inequality and frustration found in current data annotation tasks, supporting an entry-level avenue for full time data entry jobs for neurodivergent workers.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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